



Hack Away Hunger 2026

dsmHack x Corteva

Submission

Count Me In

Volunteer hours, verified — SNAP kept

80 h

Monthly rule

8

Languages

1 sec

To verify

\$303k

Aid preserved / yr

Live app	https://count-me-in-iowa.netlify.app
Live demo stage	https://count-me-in-iowa.netlify.app/stage
Verify an attestation	https://count-me-in-iowa.netlify.app/verify (no key needed)
Source code (MIT)	https://github.com/chessoren/hackawayhunger
Team	Oren Vidal
Prizes	Grand Prize · Community Impact Award · People's Choice Award

We don't create food. We stop it from disappearing.

Submission at a glance

Volunteering → verified hours → SNAP kept. An SMS & voice assistant in 8 languages helps Iowans meet the new 80-hour rule, and gives pantries the regular volunteers they need.

The problem

Since the July 2025 federal law (P.L. 119-21), adults 18–64 without a child under 14 must prove 80 hours a month of work, training or volunteering to keep SNAP; exemptions for veterans, people experiencing homelessness and former foster youth were removed. Iowa SNAP recipients fell 9% in a year (271,880 → 247,907) while 1 in 8 Iowans face food insecurity. The law already counts volunteering hour for hour — **what's missing is the proof.**

What we built

Person

SMS / voice / web assistant in 8 languages: letter photo → 3-sentence summary + deadline; screening decided by a cited, deterministic rules engine; 80-hour gauge; 3 accessible missions; QR or 4-digit check-in; consent-gated sending; DELETE by SMS.

Host site

10-minute onboarding, QR poster, first name only, one-tap validation signed with Ed25519, grant-ready report, audit log.

State

Monthly PDF attestation with QR; verified in one second; any modification breaks the signature.

Community

Anonymized dashboard ($k \geq 10$), map of opportunity deserts for 99 counties, editable impact model ($\approx$ \$303,000/year for 500 people).

How to test in 2 minutes

1. Open **count-me-in-iowa.netlify.app/stage** and paste a Claude or OpenAI key — it stays in your browser.
2. Text **Hi** (or Hola, Jambo), then tap **Send the sample State letter**.
3. Answer in your own words; pick mission 1, 2 or 3.
4. Demo clock: **Shift starts** → **Scan the QR at the door** → **Shift ends** → scan again → **Confirm**.
5. Text **ATTESTATION**, open it, then visit **/dashboard**.

API keys — temporary.

For the submission, the app runs on each user's own AI key; no key is shipped in the code. **If Count Me In wins, the prize funds the AI, SMS and voice keys so the service runs for everyone.** The demo video was recorded with pre-written AI responses; the live app uses the real APIs. Sites, people and the sample letter are illustrative.



Count Me In

Volunteer hours, verified — SNAP kept

• Live in Iowa

80-hour rule

8 languages

80 h

🕒 Monthly rule

1 sec

🛡️ To verify

3

👥 Winners per hour



Count Me In

(515) 800-8080 · English



Hi

Hi! I'm Count Me In. I help you keep your food assistance (SNAP) under the new 80-hour rule — for free, by text or by phone call.

Private by design: I only keep your first name, phone, language and hours. Nothing goes to the State without your YES. Reply DELETE anytime to erase everything, or HELP to talk to a person. I speak: English · Español · Kiswahili · العربية · Tiếng Việt · Français · Ikinyarwanda.

What's your first name?



Text message



Send the sample State letter



STATE DEPARTMENT OF HUMAN SERVICES
Food Assistance Program - Field Office 77
P.O. Box 0000, Des Moines, IA 50300

SAMPLE
NOTICE

Date of notice: **October 1, 2026**
Case number: XX-XXXXXX

LINDA M.
1400 E Walnut St Apt 2
Des Moines, IA 50316

NOTICE: ABLE-BODIED ADULT WORK REQUIREMENT

Federal law (Public Law 119-21) changed the rules for Food Assistance (SNAP). Our records show you are between 18 and 64 years old and you do not have a child under 14 in your food assistance household.

Beginning with the benefit month of **November 2026**, you must work, volunteer, or take part in a work or training program for at least **80 hours each month**, unless you qualify for an exemption.

WHAT YOU MUST DO: Within **21 days of the date of this notice**, report your hours or ask for an exemption. Return the enclosed form 470-XXXX with proof of hours worked or volunteered.

If you do not meet the requirement, you can receive food assistance for only **3 months in a 36-month period**.

You may be exempt if you are medically certified as physically or mentally unfit for work, pregnant, or caring for an incapacitated person. Volunteer work with a public or nonprofit organization counts toward the 80 hours when it is verified.

Questions? Call the Contact Center at **1-800-000-0000**, Monday to Friday, 8:00 a.m. to 5:00 p.m. You have the right to appeal this decision.

SAMPLE DOCUMENT FOR DEMONSTRATION — Count Me In, Hack Away Hunger 2026. Not an official notice.

Meet Linda

Linda, 58

Des Moines · cleans houses 10 h a week

• Gets SNAP

No car

80 h

📅 Now required

21 days

🕒 To respond

A two-page letter in administrative English. Without help, by spring she is in the line at a pantry that already breaks records.

What changed in 2025

A federal rule just reached hundreds of thousands of lowans



18 → 64

The 80-hour rule now covers adults up to 64



Child under 14

Parents of teenagers are no longer exempt



Exemptions removed

Veterans, homelessness, former foster youth



3 months in 36

Without 80 h or an exemption, SNAP stops

-9%

Iowa SNAP recipients in one year
271,880 → 247,907

1 in 8

Iowans food insecure
73M-meal gap

The real problem

Not willingness. Proof.

01

They don't know if they're covered

or already exempt

02

They can't read the letter

long, English, tight deadline

03

They can't reach hours

no car, irregular shifts

04

They can't prove what they do

volunteering leaves no trace

05

They miss the deadline

one missed report cuts the aid



The other half

Pantries chronically lack **regular weekday volunteers** — and track attendance on paper they can't certify.

SNAP provides ~9 meals for every 1 from food banks. **Every person who loses SNAP lands in the pantry line.**

• The insight • 7 CFR 273.24

The law already counts volunteering. We built the missing proof.



The person

Keeps SNAP — no app, no password

Keeps food aid



The pantry

Gains regular volunteers

Gains volunteers



The State

Gets clean, signed proof

Clean proof

One hour of volunteering serves three people at once.



Count Me In

(515) 800-8080 · English



📄 SNAP work requirement notice —
State Department of Human Services
(sample)

- Starting in November, you must do 80 hours a month of work, volunteering or training to keep your food assistance.
- If you do not, your food assistance can stop after 3 months.
- Volunteering at a nonprofit counts, if your hours are verified.

🕒 Deadline: Thu, Oct 22

✅ Report your hours or ask for an exemption by October 22, with the form and proof of your hours.

📞 1-800-000-0000

I set a reminder 3 days before the deadline.

Let's check together if this rule applies to you. What's your first name?



Text message



Send the sample State letter



01

The letter, decoded

Snap a photo. Get 3 sentences.

Text "Hi" to a number — SMS, voice or web. Photograph the State letter, even crooked. Back, in her language: **what it says, the deadline, the one thing to do**, and a reminder 3 days before.

3

📄 Sentences

8

🌐 Languages

0

📱 Apps

02

Does this apply to me?

6 questions. A cited rule.

She answers in her own words — *"My knees hurt a bit but I can still work."* The AI turns words into facts; a **deterministic rules engine decides** and cites the regulation. Unsure? A human navigator.

1

Question at a time

0

AI decisions

1 file

Per state



Count Me In

(515) 800-8080 · English



About how many hours a week do you work for pay? (Send 0 if none.)

I clean houses, about 10 hours a week

The 80-hour rule applies to you. Don't worry: work, training and volunteering all count, in any mix. I will help you get there.

Rule cited

7 CFR 273.24 — Time limit for able-bodied adults (work = paid, in-kind, unpaid or volunteer work)

Rules file IA 2026.10-hackathon

Your paid work gives about 43 hours a month. Do you have training or English class hours this month? Send the number (0 if none).

0



Text message



Send the sample State letter





Count Me In

(515) 800-8080 · English



example: "weekday mornings" or
"Tuesdays after 2pm".

I am free anytime

I clean houses Monday and Wednesday
mornings

Here are missions that fit you:

1 Food distribution — Grace Church
Community Grocery

Tue, Oct 13, 09:00–13:00 (4 h) · 12 min
walk

2 Food distribution — Eastside
Neighborhood Pantry

Sat, Oct 10, 08:00–12:00 (4 h) · 19 min
walk

3 Welcome desk — Downtown
Community Kitchen

Sun, Oct 11, 12:00–15:00 (3 h) · Route 4
— E 14th St, about 33 min · seated work
Reply 1, 2 or 3.

1

2

3



Text message



Send the sample State letter



03

Three missions, never more

Ranked by real access.

A 12-minute walk or a DART bus line, around her cleaning schedule, optimized to reach **80 hours before the deadline**. She replies "1". At the door: scan the QR, or text a 4-digit code from any basic phone.

3

✓ Choices max

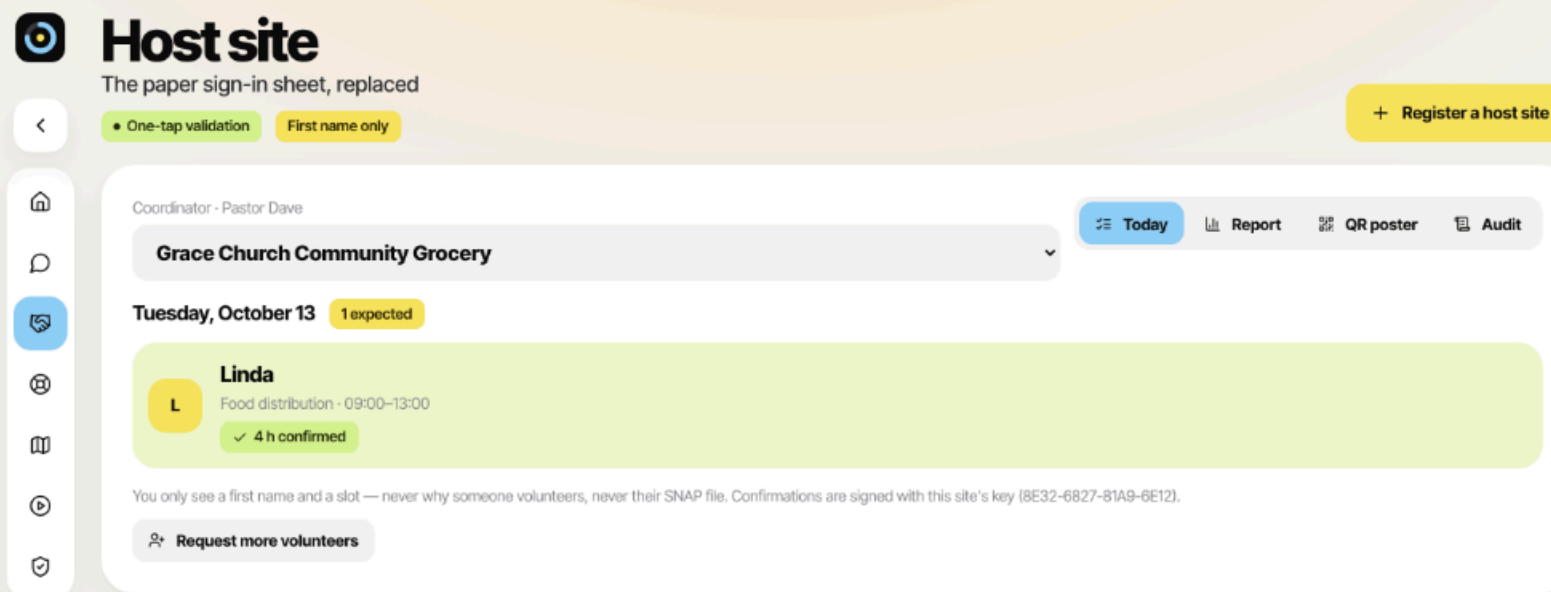
12 min

🚶 Walk

4 digits

☎ Basic phones

The paper sign-in sheet, replaced



10 min
to become a host site



One tap
confirmation, Ed25519 signed



First name
never the SNAP file



Grant-ready
monthly report & CSV

Verified in one second. Forged in none.



Verify an attestation

For caseworkers · one second · nothing stored

• No account • No AI key

Authentic

Not modified since it was signed by Count Me In.

Linda 2026-10 47 / 80 h
Name Month Total hours

Date	Organization	Mission	In-Out	Hours	Validated by
2026-10-13	Grace Church Community Grocery	Food distribution	09:02-12:59	4 h	Pastor Dave key B:32-6827-81A9-6E12
2026-10-01	Declared by the person	Paid work (self-declared, not verified by Count Me In)	—	43 h	—

Volunteering 4 h · Training 0 h · Paid work declared by the person 43 h · ID CMI-202610-02606C · issued 10/13/2026, 12:59:03 PM · rules 1A 202610-hackathon · platform key E/26-FAB0-9CC3-7F5E

Download the PDF

Verify another document

• **Authentic** signed by the platform key



Verify an attestation

For caseworkers · one second · nothing stored

• No account • No AI key

Not valid

Signature does not match the document. It was modified after signing, or not signed by Count Me In.

Linda 2026-10 51 / 80 h
Name Month Total hours

Date	Organization	Mission	In-Out	Hours	Validated by
2026-10-13	Grace Church Community Grocery	Food distribution	09:02-12:59	8 h	Pastor Dave key B:32-6827-81A9-6E12
2026-10-01	Declared by the person	Paid work (self-declared, not verified by Count Me In)	—	43 h	—

Volunteering 8 h · Training 0 h · Paid work declared by the person 43 h · ID CMI-202610-02606C · issued 10/13/2026, 12:59:03 PM · rules 1A 202610-hackathon · platform key E/26-FAB0-9CC3-7F5E

Verify another document

The attestation is signed with the Count Me In platform E825519 key 1E/26-FAB0-9CC3-7F5E. Each line references a coordinator-signed ledger entry. Changing any

+4 hours forged signature breaks instantly



Impact & opportunity deserts

lowa · 99 counties · anonymized, k ≥ 10

<10

People supported (live)

<10

Recognized as exempt

<10

Covered, on a plan

4

Volunteer hours verified

Where 80 hours are out of reach without a car

Opportunity desert · 64

Tight · 18

Enough hours · 17

64

Deserts

64 of 99 counties

are opportunity deserts: car-free accessible volunteer hours cover less than half of what people subject to the rule need. Tap a county.

County data (CSV)

Who uses this map



Food banks & pantries

Where to open the next host site



Funders & companies

Measurable impact in meals and dollars



Officials & Iowa HHS

Counties under strain, failure reasons



Researchers & press

Aggregated, downloadable data

Model: SNAP participants = population × Iowa rate (247,907 recipients, Iowa HHS May 2026); subject = 14% of participants; 35% need ~20 volunteer h/month. Supply: ~1 volunteer-hosting organization per 1,200 residents × 150 h/month × car-free share (60% with fixed-route transit, 40% mid-size, 25% small rural counties), plus registered host sites. The pilot replaces the baseline with the live site registry and transit data.

06 · For Iowa

64

of 99 counties

are **opportunity deserts**: car-free volunteer hours cover less than half of what people under the rule need.

Neutral data — where to open the next host site, or where to ask for a waiver. Never a cell under 10 people.



Count Me In

Hack Away Hunger 2026 · dsmHack × Corteva

Food aid preserved per year

\$303,318

for 500 people supported · every number editable on the live dashboard

$500 \times 40\%$ would lose SNAP $\times 75\%$ kept = **150 people**

$150 \times \$168.51$ Iowa average benefit $\times 12$ months

89,211

meals / year

36,000

volunteer hours / year

~\$5k

running cost / year

How it's built

AI understands. Rules decide. A human settles doubts.

The person

SMS · voice · web

The coordinator

QR · one-tap validation

Public & officials

Dashboard · map

Multilingual AI agent

Claude / OpenAI · structured outputs

Letter reader

Vision → summary + deadline

Rules engine

One YAML per state, cited

Matching

Walk · DART · schedule

Signed hours ledger

Hash chain · Ed25519 · audit

Human relay

Navigator console

Verifiable attestation

PDF + QR

Iowa HHS

Only after a YES

Anonymized data

$k \geq 10$ by county



Count Me In

(515) 800-8080 · Kiswahili



Jambo

Habari! Mimi ni Count Me In. Ninakusaidia kubaki na msaada wako wa chakula (SNAP) chini ya sheria mpya ya saa 80 — bure, kwa ujumbe au kwa simu.

Faragha kwanza: ninahifadhi jina lako la kwanza, simu, lugha na saa tu. Hakuna kinachoenda kwa Serikali bila YES (NDIYO) yako. Jibu DELETE (FUTA) wakati wowote kufuta kila kitu, au HELP (MSAADA) kuongea na mtu.
Ninazungumza: English · Español · Kiswahili · العربية · Tiếng Việt · Français · Ikinyarwanda.

Jina lako la kwanza ni nani?



Text message



Send the sample State letter



Dignity · access · privacy

Any phone. Any language. Always a human.



8 languages + voice for people who read little



Any phone keywords & 4-digit codes work without AI



HELP, always a human navigator one word away



DELETE by SMS minimal data, pseudonymous ledger



One option never a condition to receive food

What's next

Oct 2026

MVP

Full SMS journey, QR, attestation, map

Nov 26 – Mar 27

DMARC pilot

3–5 host sites · English + Spanish

2027

All of Iowa

55 counties · 8 languages · HHS link

2028

Other states

One rules file per state · open source

We don't create food. **We stop it** from disappearing.



count-me-in-iowa.netlify.app

>>

Judging runs on your own AI key (temporary). If we win, the prize funds the keys for everyone.

Inspiration

I'm sixteen. I have never received a letter from a government agency that could take away my food. Linda has.

Linda is 58, lives in Des Moines, cleans houses ten hours a week and has no car. In October a letter tells her that, under a new federal law, she must now show **80 hours a month** of work, training or volunteering, or lose her SNAP food assistance. She has 21 days.

Linda is a composite, but the numbers behind her are real. The 2025 law (P.L. 119-21) extended SNAP's work rule from age 54 to 64, kept the parent exemption only for households with a child under 14, and removed the exemptions for veterans, people experiencing homelessness and former foster youth. In one year, Iowa's SNAP rolls fell **9%** (271,880 → 247,907), while **1 in 8 Iowans** face food insecurity. Everyone who loses SNAP ends up in a pantry line that is already breaking records.

Then I found the sentence that became the project, in the regulation itself (**7 CFR 273.24**): volunteering with a community organization **already counts toward the 80 hours, hour for hour**, as long as the hours are verified.

So the law already gives Linda a way to keep her food. What it doesn't give her is a way to prove it. People don't lose SNAP because they refuse to work. They lose it because they can't prove what they do. And the pantries nearby are short of regular volunteers and still use paper sign-in sheets.

One well-recorded hour helps three people at once: **Linda keeps her food, the pantry gains a volunteer, and the State gets clean proof.**

I can't vote on this law, and I don't run a food bank. But I could build the missing piece, and I wanted to build it for Linda first: a tired person, on a cheap phone, maybe in her second language.

What it does

Count Me In turns volunteering into verified hours.

For the person, on any phone, in 8 languages. No app, no account. She texts a number, calls it, or uses the web.

- **The letter, decoded:** she photographs the letter and gets back three plain sentences, the deadline and the one thing to do, with an automatic reminder.
- **"Does this apply to me?":** six questions, one at a time, answered in her own words. A deterministic rules engine, not the AI, decides between "probably exempt", "covered" and "let's check with a human", and cites the regulation.
- **A plan:** an 80-hour gauge, then **three missions at most**, ranked by real access: walking time, bus line, schedule, language.
- **Proof:** she checks in with a **QR code**, or a **4-digit code** texted from any basic phone.
- **Control:** HELP reaches a human. DELETE erases everything.

For host sites. Pantries, churches and volunteer programs sign up in about ten minutes and print a QR poster. The coordinator sees only a first name and a time slot, never a SNAP file, and confirms hours in one tap. Each confirmation is cryptographically signed.

For the State. Every month, Linda gets an attestation with a QR code. A caseworker scans it and knows in one second that it's authentic. **Change a single digit and the signature breaks.** Nothing is sent to Iowa HHS without her explicit YES.

For Iowa. An anonymized dashboard and a **map of opportunity deserts**: for each of the 99 counties, how many volunteer hours are reachable without a car compared with how many are needed. Under our model, 64 counties fall short, most of them rural.

Impact you can check in your head: 500 people supported $\times$ 40% who would lose SNAP $\times$ 75% who keep it with the tool = 150 people. $150 \times \$168.51$ (Iowa's real average benefit) $\times 12 \approx$ **\$303,000 of food assistance preserved per year**, for a running cost of a few thousand dollars.

■ How we built it

I built Count Me In with an AI coding agent. My part was the research, the architecture and the decisions, and saying "no, that's not it" until it was.

- **React + TypeScript + Vite**, mobile-first and accessible, with a voice mode for people who read little.
- **One agent for every channel:** the same code answers the web phone and **Twilio** SMS and voice. Keywords and site codes work without any AI, so basic phones and outages never break the flow.
- **AI layer:** Claude or OpenAI, always with **structured JSON outputs**. It only understands free text, reads letter photos and translates.
- **Rules engine:** `rules/iowa.yaml`, one readable file per state, with its sources. When the law changes, you edit the file, not the code.
- **Signed ledger:** append-only, hash-chained, with Ed25519 signatures. It stores a pseudonym, never a phone number.
- **Attestations:** signed on the server, rendered as a PDF with a QR code and a verification page.
- **Quality:** 20 automated tests, including forgery and tampering cases, and a full end-to-end run. Hosted on **Netlify**.

About the API keys (temporary). For judging, the app asks for your own Claude or OpenAI key. It stays in your browser and is never in the code. **If we win, the prize will fund the AI, SMS and voice keys so the service runs for everyone.** Linda will never see a key; she'll just text a number. Because I shipped no keys, the demo video uses pre-written AI responses; the live app uses the real APIs.

■ Challenges we ran into

- **Getting the law exactly right.** Sources contradicted each other, so I only encoded rules I could cite.
- **Keeping the AI out of decisions without making it cold.** The AI turns words into facts; the rules engine decides. Making that still sound kind took many rounds.
- **Translating without breaking SMS.** Option numbers, codes and keywords must survive every language.
- **Making proof hard to fake.** I wrote tests that try to forge an attestation and made sure they all fail.

- **Starting over on design.** The first version worked but looked generic, so I rebuilt every screen around a single design system.

■ Accomplishments that we're proud of

A complete journey you can test in two minutes, from "Hi" to a signed attestation a caseworker can trust in one second. It works on any phone, in eight languages, with a human always one word away. And the idea it's built on: **the problem is proof, not willingness.**

■ What we learned

The bottleneck is verification, not willingness, and paperwork is something software is good at. "AI-first" doesn't have to mean "AI decides": a cited rules engine makes the tool easier to trust. Dignity has to be designed in, from the words we use to sending nothing without a yes. And being sixteen is not a reason to build something small.

■ What's next for Count Me In

1. **A pilot in the DMARC pantry network**, with 3 to 5 host sites, in English and Spanish first.
2. **Validating the attestation format with Iowa HHS.**
3. **Scaling to the rest of Iowa**, then to other states, one rules file each.

The code is open source (MIT), so a nonprofit can host it and dsmHack volunteers can maintain it. I'd love Corteva to be the first corporate host site.

I can't change the law. I can make sure nobody loses their food because they couldn't prove an hour they actually gave.

We don't create food. We stop it from disappearing.

Demo host sites, people and the sample letter are illustrative; no partnership is implied.